

## Celestra cheatsheet – v7.1.0.1 – <https://github.com/Serrin/Celestra/>

| Core API                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                   |                                                                                                                            | DOM API                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
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| constant(value);<br>identity(value);<br>noop();<br>T();<br>F();                                                                                                                                                                                                                                                                                                                         | asyncConstant(value);<br>asyncIdentity(value);<br>asyncNoop();<br>asyncT();<br>asyncF();                                                                                                                                                                                          | eq(value1, value2);<br>gt(value1, value2);<br>gte(value1, value2);<br>lt(value1, value2);<br>lte(value1, value2);          | qsa(selector[,context]);<br>qs(selector[,context]);<br>domReady(callback);<br>domClear(element);<br>domCreate(type[,properties[,innerHTML]]);<br>domCreate(element descriptive object);<br>domToElement(htmlString);<br>domGetCSS(element[,property]);<br>domSetCSS(element,property,value);<br>domSetCSS(element,properties);<br>domFadeIn(element[,duration[,display]]);<br>domFadeOut(element[,duration]);<br>domFadeToggle(element[,duration[,display]]);<br>domShow(element[,display]);<br>domHide(element);<br>domToggle(element[,display]);<br>domIsHidden(element);<br>domScrollToTop();<br>domScrollToBottom();<br>domScrollToElement(element[,top=true]);<br>domSiblings(element);<br>domSiblingsPrev(element);<br>domSiblingsLeft(element);<br>domSiblingsNext(element);<br>domSiblingsRight(element);<br>domGetCSSVar(name);<br>domSetCSSVar(name,value);<br>importScript(script1[,scriptN]);<br>importStyle(style1[,styleN]);<br>setFullscreenOn(selector);<br>setFullscreenOn(element);<br>setFullscreenOff();<br>getFullscreen();<br>form2array(form);<br>form2string(form);<br>getDoNotTrack();<br>getLocation(success[,error]);<br>createFile(filename,content[,dType]); |
| VERSION;<br><br>BASE16 = "0123456789ABCDEF";<br>BASE32 = "23456789ABCDEFGHIJKLMNPOQRSTUVWXYZ";<br>BASE36 =<br>"0123456789ABCDEFGHIJKLMNPOQRSTUVWXYZ";<br>BASE58 =<br>"123456789ABCDEFGHIJKLMNPQRSTUVWXYZabcdefghijklmnopqrstu<br>vwxyz";<br>BASE62 =<br>"0123456789ABCDEFGHIJKLMNPOQRSTUVWXYZabcdefghijklmnopq<br>rstuvwxyz";<br>WORDSAFEALPHABET =<br>"23456789CFGHJMPQRVWXCfghjmqvw"; | delay(milisec).then(callback);<br><br>bind(function,context);<br>unBind(function);<br><br>curry(function);<br>once(function);<br>tap(function): function(value);<br>compose(function1[,functionN]);<br>pipe(function1[,functionN]);<br><br>assert(condition[,message <br>error]); | RandomBoolean();<br>randomUUIDv7(v4=false);<br>timestampID([size=21[,alphabet=<br>"23456789CFGHJMPQRVWXCfghjmqvw<br>x"]]); |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| deepAssign(target,source1[,sourceN]);<br>sizeIn(object);<br>pick(object,keys);<br>omit(object,keys);<br>assoc(object,key,value);                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                   |                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| String API                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                   |                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| b64Decode(string);<br>b64Encode(string);<br><br>strCodePoints(string);<br>strFromCodePoints(iterator);<br><br>strAt(string,index[,newChar]);<br>strCount(string, substring);<br>strSplice(string,index,count[,add]);<br>strTruncate(string);<br>strReverse(string);                                                                                                                     | strUpFirst(string);<br>strDownFirst(string);<br><br>strCapitalize(string);<br>strPropercase(string);<br>strTitlecase(string);<br><br>strHTMLRemoveTags(string);<br>strHTMLEscape(string);<br>strHTMLUnEscape(string);                                                             |                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |

| Collections API                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Math API                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Type API                                                                                                                                                                                                                             |                         |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| castArray(value);<br>arrayDeepClone(array);<br>arrayMerge(target, source1[, sourceN]);<br>arrayAdd(array, value);<br>arrayClear(array);<br>arrayRemove(array, value[, all=false]);<br>arrayRemoveBy(array, callback[, all=false]);<br>arrayRange([start=0[, end=99[, step=1]]]);<br>arrayCycle(iterator[, n=100]);<br>arrayRepeat(value[, n=100]);<br>iterRange([start=0[, step=1[, end=Infinity]]]);<br>iterCycle(iterator[, n=Infinity]);<br>iterRepeat(value[, n=Infinity]);<br>(iterator, callback);<br>findLast(iterator, callback);<br>zip(iterator1[, iteratorN]);<br>unzip(iterator);<br>zipObj(iterator1, iterator2);<br>sort(iterator[, numbers = false]);<br>concat(iterator1[, iteratorN]);<br>join(iterator[, separator=","]);<br>slice(iterator[, begin=0[, end=Infinity]]);<br>unique(iterator[, resolver]);<br>includes(collection, val[, comparator]);<br>item(iterator, index);<br>nth(iterator, index);<br>first(iterator);<br>head(iterator);<br>last(iterator);<br>reverse(iterator);<br>initial(iterator);<br>tail(iterator);<br>shuffle(iterator);<br>compact(iterator);<br>size(collection);<br>min(value1[, valueN]);<br>max(value1[, valueN]); | sum(value1[, valueN]);<br>avg(value1[, valueN]);<br>product(value1[, valN]);<br><br>clamp(value, min, max);<br>minmax(value, min, max);<br><br>inRange(value, min, max);<br><br>signbit(value);<br><br>randomInt([max]);<br>randomInt(min, max);<br><br>randomFloat([max]);<br>randomFloat(min, max);<br><br>add(value1, value2);<br>sub(value1, value2);<br>mul(value1, value2);<br>div(value1, value2);<br><br>divMod(value1, value2);<br>mod(value1, value2);<br><br>pow(base, power);<br><br>isEven(value);<br>isOdd(value);<br><br>isFloat(value);<br><br>toInteger(value);<br>toIntegerOrInfinity(value); | isPropertyKey(value);                                                                                                                                                                                                                | toPropertyKey(value);   |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isIndex(value);                                                                                                                                                                                                                      | toIndex(value);         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isIterator(value);                                                                                                                                                                                                                   | isAsyncIterator(value); |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isIterable(value);                                                                                                                                                                                                                   | isAsyncIterable(value); |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | typeof(value);<br>constructorOf(value);<br>is(value, expectedType);<br>isEmpty(value);<br>isTypedCollection(iter, expectedType, Throw=false);<br>isSameType(value1, value2[, type]);<br>isSameInstance(value1, value2, Constructor); |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isDeepStrictEqual(value1, value2);<br>toSafeString(value);                                                                                                                                                                           |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isNull(value);<br>isUndefined(value);<br>isNullish(value);<br>isNonNullable(value);<br>isNonNullablePrimitive(value);<br>isPrimitive(value);                                                                                         |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isObject(value);<br>isPlainObject(value);<br>isCoercedObject(object);<br>isArraylike(value);                                                                                                                                         |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isTypedArray(value);<br>isProxy(value);<br>isElement(value);<br>isRegexp(value);                                                                                                                                                     |                         |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | isFunction(value);<br>isGeneratorFunction(value);<br>isAsyncFunction(value);<br>isAsyncGeneratorFunction(value);<br>isArrowFunction(value);                                                                                          |                         |

| How to import                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                              |            |
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| Celestra for Node.js and Deno: <i>celestra.js</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Celestra for browser: <i>celestra.js</i>                                                                                                                                                                                                                                                                                                                                     |            |
| <pre>// import the defaultExport object import defaultExport from "./celestra.js"; globalThis.celestra = defaultExport; globalThis.CEL = defaultExport;  // import with default with name import { default as celestra } from "./celestra.js"; globalThis.celestra = celestra; globalThis.CEL = celestra;  // import all into a new celestra object // includes the DOM API and Cookie API import * as celestra from "./celestra.js"; globalThis.celestra = celestra; globalThis.CEL = celestra;  // import some functions import { first, last } from "./celestra.js"; globalThis.first = first; globalThis.last = last;</pre> | <pre>&lt;script type="module"&gt; // import all into a new celestra object import * as celestra from "./celestra.js"; globalThis.celestra = celestra; globalThis.CEL = celestra; &lt;/script&gt;  &lt;script type="module"&gt; // import some functions import { first, last } from "./celestra.js"; globalThis.first = first; globalThis.last = last; &lt;/script&gt;</pre> |            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Removed APIs in the <i>defaultExport</i>                                                                                                                                                                                                                                                                                                                                     |            |
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | DOM API                                                                                                                                                                                                                                                                                                                                                                      | Cookie API |

| Cookie API                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Polyfills                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <pre>getCookie([name]); hasCookie(name);  setCookie(name,value[,hours=8760[,path="/"[,domain[,secure[,SameSite="Lax"[,HttpOnly]]]]]]); setCookie(Options object: properties are the same as the parameters);  removeCookie(name[,path="/"[,domain[,secure[,SameSite="Lax"[,HttpOnly]]]]]); removeCookie(Options object: properties are the same as the parameters); clearCookies([path="/"[,domain[,sec[,SameSite="Lax"[,HttpOnly]]]]]); clearCookies(Options object: properties are the same as the parameters);</pre> | <pre><u>globalThis.AsyncFunction()</u>; <u>globalThis.AsyncGeneratorFunction()</u>; <u>globalThis.GeneratorFunction()</u>;  <u>crypto.randomUUID()</u>;  <u>Error.isError()</u>;  <u>globalThis</u>;  <u>Math.sumPrecise()</u>;</pre> |

| Removed Polyfills - Available in celestra-polyfills.dev.js and celestra-polyfills.min.js                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |        |        |
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| v3.1.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | v3.8.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | v5.6.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |        |        |
| Array.from();<br>Array.of();<br>Array.prototype.copyWithin();<br>Array.prototype.fill();<br>Array.prototype.find();<br>Array.prototype.findIndex();<br>Object.create();<br>String.fromCodePoint();<br>String.prototype.codePointAt();<br>String.prototype.endsWith();<br>String.prototype.startsWith();<br>Math.acosh();<br>Math.asinh();<br>Math.atanh();<br>Math.cbrt();<br>Math.clz32();<br>Math.cosh();<br>Math.expml();<br>Math.fround();<br>Math.hypot();<br>Math.imul();<br>Math.log1p();<br>Math.log10();<br>Math.log2();<br>Math.sign();<br>Math.sinh();<br>Math.tanh();<br>Math.trunc();<br>Number.EPSILON;<br>Number.isNaN();<br>isNaN();<br>Number.isInteger();<br>Number.isFinite();<br>Number.isSafeInteger();<br>Number.parseInt();<br>Number.parseFloat(); | Array.prototype.values();<br>Array.prototype.includes();<br><br>ChildNode.after();<br>ChildNode.before();<br>ChildNode.remove();<br>ChildNode.replaceWith();<br><br>Element.prototype.closest();<br>Element.prototype.getAttributeNames();<br>Element.prototype.matches();<br>Element.prototype.toggleAttribute();<br><br>ParentNode.append();<br>ParentNode.prepend();<br><br>String.prototype[Symbol.iterator]();<br>String.prototype.includes();<br>String.prototype.repeat();<br><br>NodeList.prototype.forEach();<br><br>Object.assign();<br>Object.entries();<br>Object.getOwnPropertyDescriptors();<br>Object.values();<br><br>RegExp.prototype.flags;<br><br>window.screenLeft;<br>window.screenTop; | Array.prototype.at();<br>Array.prototype.findLast();<br>Array.prototype.findLastIndex();<br>Array.prototype.flat();<br>Array.prototype.flatMap();<br>Number.MIN_SAFE_INTEGER;<br>Number.MAX_SAFE_INTEGER;<br>Object.fromEntries();<br>Object.is();<br>String.prototype.at();<br>String.prototype.matchAll();<br>String.prototype.padStart();<br>String.prototype.padEnd();<br>String.prototype.replaceAll();<br>String.prototype.trimStart();<br>String.prototype.trimLeft();<br>String.prototype.trimEnd();<br>String.prototype.trimRight();<br>Typedarray.prototype.at();<br>TypedArray.prototype.findLast();<br>TypedArray.prototype.findLastIndex();<br><br><th>v5.9.0</th><br>BigInt.prototype.toJSON();<br><br><th>v6.5.0</th><br>Array.fromAsync();<br>Array.prototype.toReversed();<br>Array.prototype.toSorted();<br>Array.prototype.toSpliced();<br>Array.prototype.with();<br>Map.groupBy(); and Object.groupBy();<br>Object.hasOwn();<br>TypedArray.prototype.toReversed();<br>TypedArray.prototype.toSorted();<br>TypedArray.prototype.with(); | v5.9.0 | v6.5.0 |